

Forget-It-All: Multi-Concept Machine Unlearning via Concept-Aware Neuron Masking

Kaiyuan Deng¹ Bo Hui² Gen Li³ Jie Ji³ Minghai Qin⁴ Geng Yuan⁵ Xiaolong Ma¹

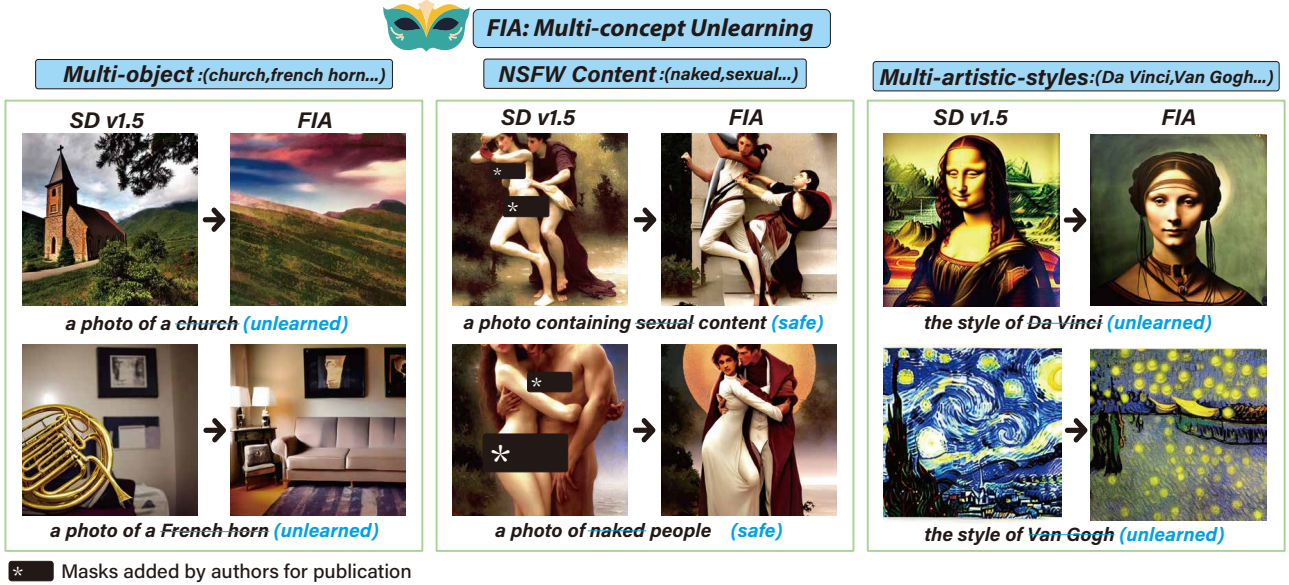


Figure 1. The proposed FIA framework enables simultaneous multi-concept unlearning in text-to-image models. We demonstrate the unlearning effects of 2 concepts in the multi-concept scenario with FIA (more comprehensive results are shown in Section 4). It shows that FIA can (i) unlearn multiple undesired objects, (ii) prevent the generation of explicit content, and (iii) mitigate artwork copyright issues. This figure illustrates that FIA not only achieves robust multi-concept unlearning but also preserves the generation quality.

Abstract

The widespread adoption of text to image (T2I) diffusion models has raised concerns about their potential to generate copyrighted, inappropriate, or sensitive imagery. As a practical solution, *machine unlearning* aims to erase unwanted concepts without retraining from scratch. While most existing methods are effective for single concept unlearning, they often struggle when removing multiple concepts, causing significant challenges in unlearning effectiveness, generation quality, and sensitivity to hyperparameters and datasets. We take a unique perspective on multi-concept unlearning by leveraging model sparsity and pro-

pose the Forget It All (FIA) framework. FIA first introduces *Contrastive Concept Saliency* to quantify each weight connection’s contribution to a target concept. It then identifies *Concept Sensitive Neurons* by combining temporal and spatial information, ensuring that only neurons consistently responsive to the target concept are selected. Finally, FIA constructs masks from the identified neurons and fuses them into a unified multi concept mask, where *Concept Agnostic Neurons* that broadly support general content generation are preserved while concept specific neurons are pruned to remove the targets. FIA is training free and requires minimal hyperparameter tuning for new tasks, enabling plug and play use. Extensive experiments across three distinct unlearning tasks demonstrate that FIA achieves more reliable multi-concept unlearning, improving forgetting effectiveness while maintaining generation fidelity and quality. Code is available at <https://github.com/kaiyuan02415/Forget-It-All>

¹The University of Arizona ²The University of Tulsa ³Clemson University ⁴Western Digital Corporation ⁵University of Georgia. Correspondence to: Kaiyuan Deng <kaiyuan0415@arizona.edu>.

1. Introduction

Text-to-image (T2I) diffusion models (Rombach et al., 2022; Ma et al., 2024; Podell et al., 2023; Saharia et al., 2022; Nichol et al., 2021; Ding et al., 2022; Zhou et al., 2022) have achieved strong performance and versatility across diverse real-world applications (Kazerouni et al., 2022; Xing et al., 2024). However, these models also pose significant ethical and legal risks (Weidinger et al., 2021; Brundage et al., 2018; Tolosana et al., 2020; Solaiman & Dennison, 2021; Xu et al., 2024; Marino et al., 2025; Liu et al., 2024), including copyright infringement and the generation of harmful content, underscoring the need for practical mitigation strategies. A promising solution is machine unlearning (Bourtole et al., 2021; Guo et al., 2019; Graves et al., 2021; Jang et al., 2022; Nguyen et al., 2022; Zhang et al., 2023; Xiao et al., 2025b), which selectively removes the influence of problematic data from pre-trained models without degrading overall performance. By preventing the generation of specified concepts, machine unlearning enhances copyright compliance and model safety while avoiding costly retraining from scratch.

Most existing machine unlearning (MU) methods (Zhang et al., 2024; Fan et al., 2023; Heng & Soh, 2023; Kumari et al., 2023; Gandikota et al., 2023; Lyu et al., 2024; Lu et al., 2024; Zhao et al., 2024; Xiao et al., 2025c) are primarily designed to erase single concepts. Fine-tuning-based methods (Zhang et al., 2024; Fan et al., 2023; Kumari et al., 2023; Gandikota et al., 2023) achieve concept erasure by updating the cross-attention layers of diffusion models. In contrast, training-free approaches, such as weight editing (Chavhan et al., 2024) and Elastic Weight Consolidation (Heng & Soh, 2023), remove concepts without fine-tuning. However, these single-concept unlearning methods struggle in real-world scenarios involving multiple concepts. When applied sequentially, they often cause the model to re-acquire forgotten concepts or suffer degraded generative quality.

To achieve multi-concept unlearning, SPM (Lyu et al., 2024) and MACE (Lu et al., 2024) employ low-capacity adapters or LoRA weight merging to erase multiple concepts. Meanwhile, ESD (Gandikota et al., 2023) removes concept combinations via an LLM-derived concept graph and adversarial feature decoupling, while SepME (Zhao et al., 2024) erases concepts through residual extraction and cross-attention nullspace decomposition. Furthermore, UCE (Gandikota et al., 2024) performs a closed-form edit of cross-attention weights to remove concepts. Despite this progress, existing multi-concept unlearning methods still face two major challenges: (i) they either degrade generative performance after unlearning or fail to fully remove all target concepts, limiting the balance between unlearning efficacy and generation quality; (ii) most methods rely on fine-tuning and are thus sensitive to hyperparameters, leading to complex tuning, increased computational cost, and a risk of overfitting.

In this paper, we propose **Forget-It-All (FIA)**, a training free framework that can simultaneously forget *arbitrary sets of concepts* while retaining the model’s generative quality (as shown in Figure 1). FIA introduces *Contrastive Concept Saliency* to quantify each weight connection’s contribution to a target concept. It identifies *Concept-Sensitive Neurons* by aggregating neuron responses across denoising timesteps and ranking them channel-wise and cross-channel, ensuring reliable association with the target concept. Finally, FIA constructs per-concept masks and applies a multi-concept fusion strategy that preserves *Concept-Agnostic Neurons*, which respond broadly across concepts to maintain generative quality, while pruning concept-specific neurons. FIA requires neither fine-tuning nor concept mapping, and operates with only a small set of easily controlled parameters. Its plug-and-play design makes it rapidly deployable across diverse unlearning tasks. We conducted comprehensive experiments on various datasets to evaluate FIA’s effectiveness in three distinct unlearning scenarios: (1) *multi-object unlearning*; (2) *multi-artist-style unlearning*; (3) *explicit content unlearning*. Our experimental results demonstrate that FIA consistently outperforms state-of-the-art approaches across a range of unlearning tasks. The key contributions of our work are threefold:

- We present a novel perspective that connects model sparsity with multi-concept unlearning, where each concept corresponds to a distinct neuron mask. To the best of our knowledge, this is the first work to explore multi-concept unlearning through unstructured neuron masking, enabling fine-grained control over concept forgetting.
- The proposed FIA multi-concept unlearning framework is training-free and works in a plug-and-play paradigm. It uncovers and exploits two distinct types of neurons: concept-sensitive neurons, which are selectively pruned to remove target concepts, and concept-agnostic neurons, which are preserved to maintain generative quality.
- We demonstrate that FIA generalizes across tasks. We conduct extensive experiments on multiple unlearning tasks, showing FIA achieves state-of-the-art unlearning performance at under 0.3% overall sparsity. Such robust performance paves the way for more regulated T2I applications.

2. Related Work

Machine Unlearning (MU) (Bourtole et al., 2021; Guo et al., 2019; Graves et al., 2021; Jang et al., 2022; Podell et al., 2023; Vatter et al., 2023) has emerged to address removal requests and data privacy concerns without requiring costly full retraining. Its goal is to erase the influence of specific data or learned concepts while preserving the model’s overall performance. In diffusion models, MU approaches fall into two categories: single-concept unlearning

and multi-concept unlearning.

Single-concept Unlearning. Finetuning-based approaches have been widely explored. FMN (Zhang et al., 2024) leverages attention re-steering for concept forgetting, while SalUn (Fan et al., 2023) modifies critical weights through gradient-based saliency. AC (Kumari et al., 2023) aligns generated images for a target concept with a broader anchor concept before ablating, and SA (Heng & Soh, 2023) adopts continual learning principles for concept erasure. ESD (Gandikota et al., 2023) fine-tunes the model using negative guidance to steer outputs away from undesired content, while MS (Jia et al., 2023) utilizes sparse training to unlearn concepts. However, all these methods are sensitive to hyperparameters and the characteristics of the training data. In contrast, training-free methods have also been proposed. SLD (Schramowski et al., 2023) injects safety guidance into latent space to suppress unwanted outputs, but latent biases can persist since weights are unchanged. ConceptPrune (Chavhan et al., 2024) removes neurons tied to an undesired concept; however, handling multiple concepts is difficult due to complex neuron interactions.

Multi-concept Unlearning. MACE (Lu et al., 2024) uses closed-form cross-attention refinement with multiple LoRA modules, while SPM (Lyu et al., 2024) offers a non-invasive erasure strategy. COGFD (Gandikota et al., 2023) leverages LLM-generated concept graphs for harmful concept elimination, SepME (Zhao et al., 2024) employs weight decoupling for targeted erasure, SCULPT (Li et al., 2025a) introduces dynamic masks with concept-aware optimization, and Stereo (Srivatsan et al., 2025) proposes a two-stage adversarially robust erasure framework. However, as the number of concepts grows, fine-tuning overhead increases, hyperparameter tuning becomes more complex, and balancing unlearning effectiveness with generative quality becomes increasingly challenging. In contrast, training-free methods such as UCE (Gandikota et al., 2024), ScaPre (Deng et al., 2026), and SPEED (Li et al., 2025b) introduce closed-form solutions to modify cross-attention weights for efficient concept erasure. However, they depend on precomputed concept embeddings to guide the edits, and direct parameter manipulation can inadvertently degrade image quality.

3. Method

In this section, we present FIA, a training-free framework for multi-concept unlearning. We begin in Section 3.1 by formulating a unified energy-based saliency and introducing *Contrastive Concept Saliency*, which quantifies each weight connection’s contribution to a target concept. Then, in Section 3.2, we describe how to identify *Concept-Sensitive Neurons* by integrating temporal sparsity and spatial sparsity, ensuring that only neurons truly responsive to the target concept are retained. Finally, in Section 3.3, we construct

per-concept masks from the identified neurons, introduce the notion of *Concept-Agnostic Neurons*, and design a fusion strategy that combines all single-concept masks into a unified multi-concept mask. This strategy explicitly preserves concept-agnostic neurons to maintain core generative capacity, while pruning only those neurons that are truly specific to the target concepts. An overview of the entire FIA pipeline is illustrated in Figure 2.

3.1. Contrastive Concept Saliency

To comprehensively quantify each weight neuron’s contribution to concept generation, we propose a unified energy saliency that simultaneously accounts for the structural capacity of weights, the activation magnitude of input features, and the effectiveness of signal transmission at each denoising step. Consider a linear layer ℓ with weight matrix $W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$, where C_{in} and C_{out} are the numbers of input and output channels. At timestep t , the activations are denoted as $X_{\ell,t} \in \mathbb{R}^{N \times C_{\text{in}}}$, where N is the total number of positions after flattening the batch and spatial or token dimensions. We let $X_{\ell,t,j} \in \mathbb{R}^N$ represent the activation vector of the j -th input channel, and the response of the i -th output channel is $Y_{\ell,t,i} = \sum_{j'=1}^{C_{\text{in}}} W_{\ell,i,j'} X_{\ell,t,j'}$. For the weight connection between input channel j and output channel i , the saliency score at step t is defined as:

$$U_{\ell,t,i,j} = |W_{\ell,i,j}| \cdot \|X_{\ell,t,j}\|_2 \cdot \frac{|\langle X_{\ell,t,j}, Y_{\ell,t,i} \rangle|}{\|X_{\ell,t,j}\|_2 \cdot \|Y_{\ell,t,i}\|_2 + \varepsilon}, \quad (1)$$

where $\|X_{\ell,t,j}\|_2$ and $\|Y_{\ell,t,i}\|_2$ are the ℓ_2 -norms of the input and output activations, $\langle \cdot, \cdot \rangle$ denotes the Euclidean inner product, and ε is a small constant to avoid numerical instability.

From a theoretical perspective, this metric serves as a training-free approximation of neuron importance based on *effective signal propagation* (Lee et al., 2018; Wang et al., 2020; Xiao et al., 2025a; Li et al., 2024). While gradient-based pruning methods typically rely on the loss-gradient-weight product, our formulation approximates the contribution to the latent representation without requiring backpropagation. Specifically, $|W_{\ell,i,j}| \cdot \|X_{\ell,t,j}\|_2$ measures the *magnitude of signal flow*, prioritizing connections combining high structural capacity with strong input activations. Crucially, the cosine similarity term penalizes neurons that activate strongly but transmit orthogonal (noise-like) signals not constructively contributing to output features. By integrating these aspects, $U_{\ell,t,i,j}$ effectively identifies neurons both structurally significant and functionally coherent for the target concept.

Building on this unified energy formulation, we further compute a saliency score to identify neurons that respond specifically to a target concept rather than to general background patterns. For each target concept, we evaluate $U_{\ell,t,i,j}$ under

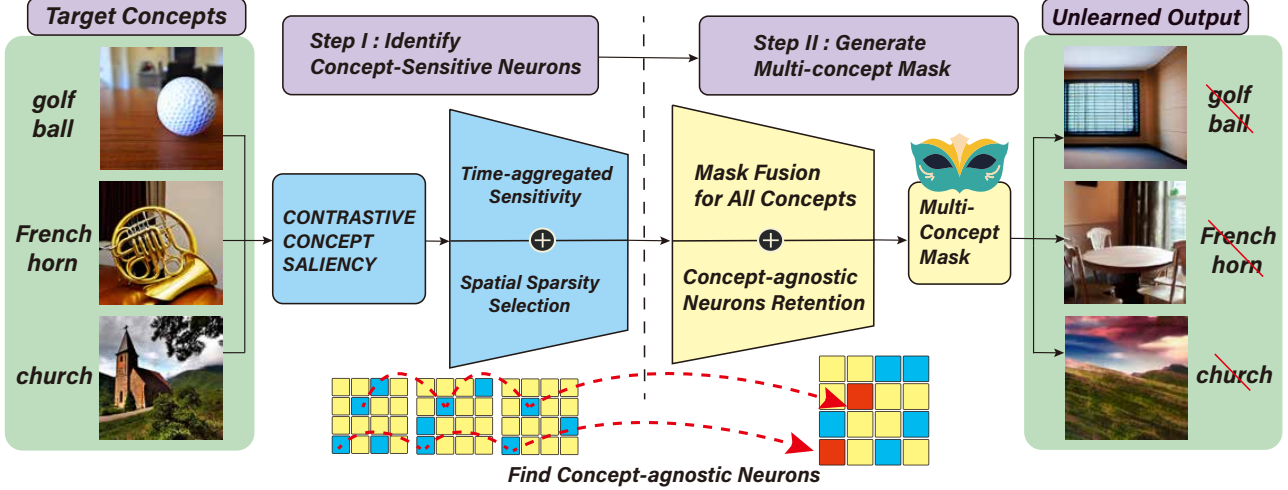


Figure 2. Overview of our unlearning framework (illustrated with golf ball, French horn, and church). We first compute *Contrastive Concept Saliency* to quantify neuron responses to target concepts. These scores are aggregated over time and refined with spatial sparsity to identify *Concept-Sensitive Neurons*. Finally, we generate per-concept masks and fuse them into a multi-concept mask while preserving *concept-agnostic neurons*.

two types of textual prompts:

- *Concept prompt*: explicitly contains the target concept (e.g., “a golf ball on the table”).
- *Base prompt*: describes only the surrounding context (e.g., “a table”).

Based on these responses, we introduce **Contrastive Concept Saliency**, which is computed as:

$$S_{\ell,t,i,j} = \max(0, \mu_c - \mu_b - \sigma_b), \quad (2)$$

where μ_c and μ_b are the mean responses of $U_{\ell,t,i,j}$ computed over K generated samples (we set $K = 50$ to ensure statistical stability) under the concept and base prompts, respectively, and σ_b is the standard deviation under the base prompts. Here, $\mu_c - \mu_b$ captures how strongly a neuron reacts to the target concept relative to the background, while subtracting σ_b filters out neurons with unstable or noisy background activations. This ensures that only neurons with a stable and statistically significant increase for the target concept are retained, making $S_{\ell,t,i,j}$ a reliable indicator of concept-sensitive neurons.

3.2. Concept-Sensitive Neurons

In this section, we introduce how to identify **Concept-Sensitive Neurons**. These neurons are those most responsive to a specific target concept across both temporal and spatial contexts, ensuring that subsequent pruning focuses only on parameters that are truly concept-related.

Time-Integrated Sensitivity. Given the per-timestep contrastive concept saliency scores $S_{\ell,t,i,j}$, we integrate temporal information to ensure robust identification of neurons across denoising steps. A direct summation across steps

would overemphasize neurons that only spike briefly, so we design a two-part sensitivity measure balancing both *response strength* and *activation persistence*. For each neuron (i, j) , the aggregated sensitivity is

$$A_{\ell,i,j} = \underbrace{\frac{1}{T} \sum_{t=1}^T S_{\ell,t,i,j}}_{\text{average response strength}} + \underbrace{\frac{1}{T} \sum_{t=1}^T \mathbf{1}[S_{\ell,t,i,j} > \tau_{\ell,t}]}_{\text{activation frequency}}, \quad (3)$$

where $\mathbf{1}[\cdot]$ is the indicator function and T is the total number of denoising steps. The first term measures how strongly the neuron responds to the concept over the whole generation process, while the second term counts the proportion of timesteps where the neuron is significantly active. The threshold $\tau_{\ell,t}$ is adaptively determined for each layer and timestep using a *temporal sparsity* r_1 , by keeping the top- r_1 fraction of neuron sensitivities:

$$\tau_{\ell,t} = \text{Top-}r_1(\{S_{\ell,t,i,j}\}_{i=1..C_{\text{out}},j=1..C_{\text{in}}}),$$

which dynamically adjusts to different noise levels and prevents a few extreme values from dominating the selection. By combining both terms with equal weight, neurons that are consistently and strongly relevant to the target concept are given higher $A_{\ell,i,j}$.

Spatial Sparsity Selection. After obtaining the time-integrated sensitivity $A_{\ell,i,j}$, we select neurons by jointly considering both channel-level and global-level information, ensuring that the most concept-relevant neurons are accurately identified.

First, for each output channel i , we rank all input neurons based on their sensitivities $A_{\ell,i,j}$ and select the top $k = r_2 \times C_{\text{in}}$ neurons with the strongest responses, where

r_2 denotes the *spatial sparsity*. This step focuses on identifying the most concept-relevant neurons for each channel individually, ensuring that the key neurons contributing to concept representation are preserved. The union of these channel-specific selections forms the local candidate set:

$$C_\ell = \bigcup_{i=1}^{C_{\text{out}}} \{(i, j) \mid j \in \text{Top}_k\{A_{\ell, i, j}\}_{j=1}^{C_{\text{in}}}\}. \quad (4)$$

Next, at the layer level, we consider all neurons across the entire layer and again rank them by their sensitivities. From this ranking, we retain the top $K_g = r_2 \times C_{\text{out}} \times C_{\text{in}}$ neurons with the highest response values. This global step filters out less relevant neurons and ensures that only the most strongly activated neurons across the whole layer are preserved, forming the global candidate set:

$$G_\ell = \{(i, j) \mid (i, j) \in \text{Top}_{K_g}\{A_{\ell, i, j}\}\}. \quad (5)$$

Finally, we integrate the two selection criteria by taking their intersection:

$$\mathcal{Q}_\ell^{(c)} = C_\ell \cap G_\ell. \quad (6)$$

This yields the final set of *Concept-Sensitive Neurons* for layer ℓ with respect to concept c , guaranteeing that the selected neurons are globally competitive while remaining evenly distributed across channels. In other words, $\mathcal{Q}_\ell^{(c)}$ captures the neurons that are most strongly associated with the target concept, both within their local context and across the entire network layer.

3.3. Multi-concept Mask Fusion

Building on the identified Concept-Sensitive Neurons $\mathcal{Q}_\ell^{(c)}$, we generate a mask for each target concept:

$$\text{Mask}_\ell^{(c)}(i, j) = \begin{cases} 1, & \text{if } (i, j) \in \mathcal{Q}_\ell^{(c)}, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

By combining temporal aggregation and spatial sparsity into one unified decision, the resulting mask highlights only neurons that are consistently active and structurally significant for representing the target concept, while unrelated neurons are marked as 0 for potential pruning. This provides a stable and precise foundation for subsequent multi-concept unlearning.

A naïve mask-fusion strategy directly fuses the masks of all concepts and prunes any neuron that is sensitive to at least one of them. In practice, this union approach severely degrades generation quality, since many neurons participate not only in specific concepts but also in core image formation and features. Empirically, we observe that a small

subset of neurons responds strongly to *most* or even *all* of the concepts we wish to forget. Such neurons clearly encode broadly useful features rather than any single concept, so we term these neurons **concept-agnostic neurons**. To identify them, we compute each neuron’s aggregate concept sensitivity (let C denote the total number of target concepts):

$$s_{\ell, i, j} = \sum_{c=1}^C \text{Mask}_\ell^{(c)}[i, j], \quad (8)$$

which counts how many of the concepts trigger a response in neuron (i, j) . We then define a *concept-agnostic threshold*: $\tau_{ca} = \lceil \alpha C \rceil$, where the **concept-agnostic ratio** $\alpha \in (0, 1]$ denotes the minimum fraction of concepts to which a neuron must be sensitive in order to be considered concept-agnostic. During mask fusion, a neuron is labeled concept-agnostic and kept if $s_{\ell, i, j} \geq \tau_{ca}$. Consequently, only truly concept-sensitive neurons with $0 < s_{\ell, i, j} < \tau_{ca}$ are pruned. By preserving these concept-agnostic neurons, we retain the model’s core generative capabilities and avoid the quality degradation caused by over-pruning. We present comprehensive ablation results in Appendix B to validate the effectiveness of preserving concept-agnostic neurons.

4. Experiment

We first describe the experimental setup (Section 4.1), followed by results on three unlearning tasks: object (Section 4.2), explicit content (Section 4.3), and artistic style unlearning (Section 4.4), with comparisons to multiple baselines. Additional experimental results and ablation studies are provided in Appendix B and F.

4.1. Experiment Setting

All experiments were conducted using Stable Diffusion v1.5. To align with prior baselines, explicit content unlearning was evaluated on Stable Diffusion v1.4. To demonstrate the *architectural generalizability* of our framework, we extend FIA to SDXL (Podell et al., 2023) in Appendix E. Multi-object unlearning is evaluated on the Imagenette benchmark (Howard et al., 2019), a ten-class ImageNet subset, while explicit content unlearning uses the I2P dataset (Schramowski et al., 2023), which includes diverse inappropriate prompts. For artistic style unlearning, we collected 200 artwork titles per artist and constructed prompts by appending the artist’s name. Model generative performance after unlearning is assessed on MS COCO-30K (Lin et al., 2014). All experiments use 50 denoising steps. For Contrastive Concept Saliency, we generate 10 samples per concept, with a sensitivity analysis provided in Appendix D. Prompt design and FIA hyperparameter settings are given in Appendix A. All experiments were run on an NVIDIA RTX A6000 GPU, with peak GPU memory usage and execution time reported in Appendix F Table 19 20.

Table 1. Forgetting accuracy ($\downarrow$) for each class under simultaneous unlearning of ten concepts on Imagenette, and CLIP score ($\uparrow$). FIA (ours) achieves the best unlearning performance.

Imagenette classes	Method									FIA (Ours)
	SD v1.5	FMN	AC	ESD	SalUn	CP	MACE	UCE	SPM	
garbage truck	89.1	78.9	49.2	31.6	10.6	6.7	82.8	28.9	53.6	0.5
cassette player	67.6	14.8	7.4	4.7	34.3	4.6	14.9	2.3	3.9	0.0
tench	98.5	79.3	11.4	64.5	92.2	0.3	84.7	5.2	43.4	0.0
English springer	98.2	85.6	92.1	79.3	1.5	2.5	93.1	0.8	67.2	1.7
chain saw	78.3	43.2	77.2	12.2	7.8	0.9	73.5	4.7	32.7	1.8
parachute	93.5	90.4	46.6	6.3	10.1	3.5	89.7	8.2	74.1	1.9
golf ball	98.2	92.7	57.2	13.1	5.9	33.2	94.6	7.8	93.8	4.8
church	86.9	77.5	82.9	61.4	1.2	8.1	69.3	19.5	66.5	0.0
French horn	98.4	87.4	94.0	57.8	9.4	2.2	96.4	3.6	17.3	2.9
gas pump	94.7	69.1	63.5	54.9	58.7	11.4	83.2	5.2	20.4	5.0
Avg Acc ($\downarrow$)	90.34	71.89	58.15	38.58	23.17	7.34	78.22	8.62	47.29	1.9
CLIP _{coco} ($\uparrow$)	31.42	30.56	31.58	30.12	29.93	27.93	31.05	29.25	30.77	29.56

Table 2. Comparison of unlearning methods on Imagenette for simultaneous unlearning of the first five concepts. Reporting forgetting accuracy ($\downarrow$) on those five classes, preservation accuracy ($\uparrow$) on the last five, and harmonic-mean based overall score. Note that we do not highlight the best preserving accuracy in bold, as high values may result from failing to forget any concepts.

Imagenette classes	Method								
	FMN	AC	ESD	SalUn	CP	MACE	UCE	SPM	<i>FIA</i> (Ours)
Classes to Forget	Forgetting Accuracy (↓)								
garbage truck	68.4	47.2	26.8	7.2	5.3	76.7	16.9	48.0	2.6
cassette player	10.1	8.4	3.4	16.9	2.8	8.3	3.1	2.6	1.4
tench	62.3	9.1	39.8	43.1	1.9	65.5	3.7	46.2	1.7
English springer	79.6	76.4	53.7	1.3	0.9	80.6	0.4	59.8	2.5
chain saw	23.9	64.2	9.5	8.1	2.8	61.3	3.6	28.2	2.4
Classes to Preserve	Preserving Accuracy (↑)								
parachute	79.2	65.1	71.0	73.3	48.3	80.7	58.6	77.1	77.2
golf ball	85.8	77.4	74.7	81.4	62.7	81.2	76.8	85.1	81.7
church	71.4	79.0	63.9	72.4	51.0	66.8	75.0	68.5	68.9
French horn	80.7	90.1	87.0	85.5	84.0	87.3	78.7	80.3	86.4
gas pump	67.4	78.9	64.4	74.2	22.5	74.9	75.2	73.0	67.9
Forgetting Acc [1–5] (↓)	48.9	41.1	26.6	22.3	2.7	58.5	5.5	37.0	2.1
Preserving Acc [6–10] (↑)	76.9	78.1	72.2	77.4	52.4	78.2	71.9	76.5	76.7
Overall Score (↑)	61.4	67.2	72.8	77.5	68.1	54.2	81.7	69.1	86.0

4.2. Multi-object Unlearning

In Table 1, we report results for simultaneously forgetting all ten Imagenette classes. Forgetting accuracy is measured using a pretrained ResNet-50 classifier (He et al., 2016), following the same evaluation protocol as prior work. FIA achieves the lowest average forgetting accuracy of 1.9%, indicating the most complete removal of target concepts. In terms of generative quality, FIA attains a CLIP score of 29.56 on MS COCO 30K (Lin et al., 2014), outperforming training-free baselines such as CP (Chavhan et al., 2024) and UCE (Gandikota et al., 2024). While several finetuning-based methods (FMN (Zhang et al., 2024), ESD (Gandikota et al., 2023), AC (Kumari et al., 2023), SalUn (Fan et al., 2023), SPM (Lyu et al., 2024), and MACE (Lu et al., 2024)) achieve slightly higher CLIP scores, they suffer from substantially higher forgetting accuracies. In contrast, FIA delivers markedly lower forgetting while maintaining competitive generative performance, demonstrating superior concept unlearning. In Table 2, we further evaluate the

trade-off between forgetting five classes and preserving the remaining five. We intentionally avoid bolding the highest preserving accuracy, as models that fail to forget may trivially retain all concepts and obtain inflated scores. Instead, we report an *Overall Score* that jointly reflects forgetting and preservation. This score is defined as the harmonic mean of preserving accuracy P and forgetting rate $R = 1 - F$, i.e. $\text{Overall Score} = 2 \frac{P(1-F)}{P+(1-F)} \times 100\%$. FIA achieves 86% on this metric, outperforming all baselines. Overall, these results confirm that FIA achieves the best balance between effective concept unlearning and preservation quality, establishing a new state of the art for multi-concept unlearning. Additional results are provided in Appendix F (Figure 7–8 and Table 21–22).

To systematically evaluate both unlearning efficacy and concept preservation, we erase x Imagenette classes and preserve the remaining ones, measuring how well each method erases target concepts while maintaining other content. As shown in Figure 4(a), FIA consistently yields the lowest

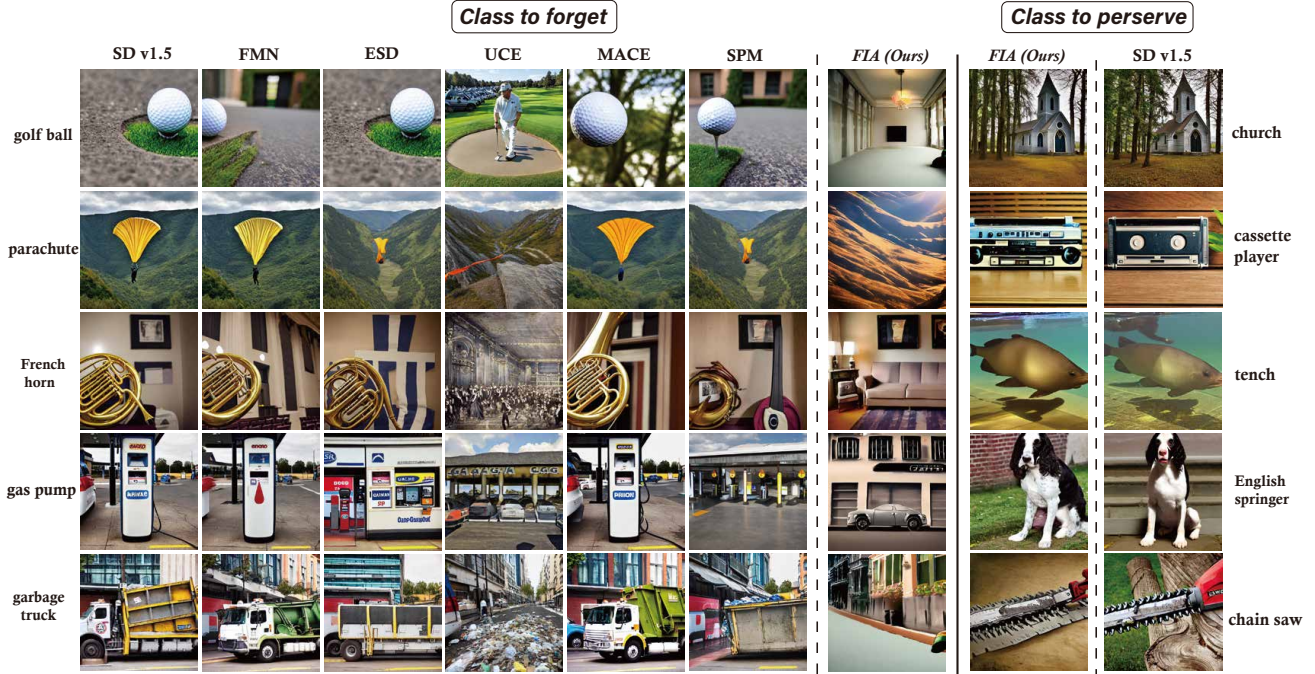


Figure 3. Visual results on the Imagenette dataset, demonstrating simultaneous unlearning of five target classes while preserving the other five. Our method achieves superior unlearning performance on the target classes, and continues to faithfully generate the preserved classes. More visual results can be found in Figure 7,8.

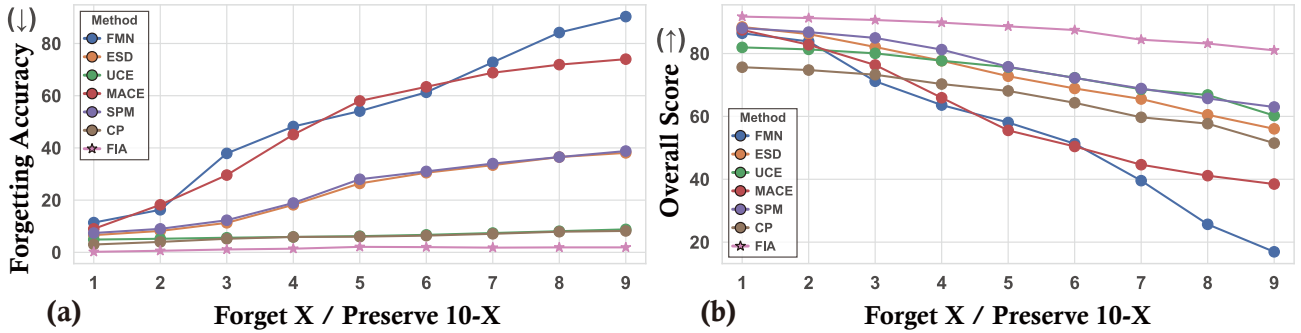


Figure 4. Forgetting accuracy (a) and Overall Score (b) on Imagenette across various forget-preserve configurations, demonstrating FIA’s superior balance between unlearning efficacy and generation quality.

forgetting accuracy for every forget-preserve configuration, outperforming all baselines even as the number of forgotten concepts increases. The overall score plotted in Figure 4(b) further demonstrates that FIA achieves the best trade-off between concept removal and generation quality. These results confirm the robustness of our approach in balancing forgetting and preservation across all settings. We also *expand multi-concept unlearning to a larger scale*, and we show the relevant results in Appendix F (Figure 6 and Table 24).

4.3. Explicit Content Unlearning

We evaluate explicit concept unlearning on the I2P (“Inappropriate Image Prompts”) benchmark, which contains 4,703 real-world text-to-image prompts that are highly prone to generate inappropriate content. Our evaluation uses three

complementary metrics: (1) NudeNet detection counts to quantify concept unlearning, (2) FID (Fréchet Inception Distance) to assess visual quality, and (3) CLIP score to measure semantic consistency. NudeNet (Bedapudi, 2019) is a lightweight nudity detector offering both classification and localization of explicit elements to identify exposed body regions (armpits, belly, buttocks, feet, female breasts, female genitalia, male breasts, and male genitalia). Following prior work, we mark a region as inappropriate only if NudeNet’s confidence exceeds 0.6, matching all baselines’ settings (Schramowski et al., 2023; Lu et al., 2024). As shown in Table 3, FIA reduces the total number of NudeNet detections on Stable Diffusion v1.4 from 743 to just 32, outperforming every competitor. On MS COCO 30K (Lin et al., 2014), FIA achieves an FID of 14.02, nearly matching

Table 3. Results of NudeNet detection result on the I2P dataset. “(F)” denotes female, “(M)” denotes male. [†] Partial results from MACE (Lu et al., 2024) and SA (Heng & Soh, 2023).

Method	NudeNet Detection									Metric	
	Armpits	Belly	Buttocks	Feet	Breasts (F)	Genitalia (F)	Breasts (M)	Genitalia (M)	Total ↓	FID ↓	CLIP ↑
FMN	47	120	23	54	163	17	21	3	448	13.54	30.43
AC	153	180	45	66	298	22	67	7	838	14.13	31.37
UCE	29	62	7	29	35	5	11	4	182	14.07	30.85
CP	36	31	7	8	49	4	4	9	148	14.11	31.04
SLD-M	47	72	3	21	39	1	26	3	212	16.34	30.90
ESD	59	73	12	39	100	6	18	8	315	14.41	30.69
SA [†]	72	77	19	25	83	16	0	0	292	–	–
SPM	51	69	8	14	70	5	10	2	229	13.81	31.24
MACE [†]	17	19	2	39	16	2	9	7	111	13.42	29.41
FIA (Ours)	6	2	7	2	6	0	1	8	32	14.02	31.18
SD v1.4	148	170	29	63	266	18	42	7	743	14.04	31.34

Table 4. Comparison of unlearning methods for simultaneous unlearning of five artist styles.

Method	Artist unlearning		MS COCO-30K		Rank ↓
	CLIP _a ↓	FSR ↑	FID ↓	CLIP ↑	
FMN	30.27	52.8	21.4	30.82	4.75
ESD	33.62	39.2	17.1	30.52	6.50
UCE	31.89	44.0	19.7	31.19	5.50
AC	33.59	45.2	16.6	31.28	4.00
CP	27.90	79.6	18.4	29.76	4.50
MACE	30.98	57.4	15.9	30.14	3.75
SPM	31.10	40.0	17.4	31.33	4.50
FIA (Ours)	27.45	83.4	16.7	30.56	2.50
SD v1.5	42.10	–	14.5	31.34	–

MACE’s best score of 13.42, and a CLIP score of 31.18, demonstrating that images after unlearning maintain high visual quality and semantic fidelity. These results establish FIA as the new state of the art for explicit concept unlearning under strict, reproducible conditions. More experimental and visual results are presented in Appendix F (Figure 9 and Table 18).

4.4. Multi-Artistic-Style Unlearning

For artistic styles unlearning, our goal is to simultaneously forget the styles of five famous artists (Van Gogh, Monet, Picasso, Da Vinci, and Dali). To evaluate unlearning effectiveness, we report the CLIP score (CLIP_a) and **Forget-Success Rate** (FSR). For each prompt we generate paired images with the original SD model and with the edited model, using the same random seed. we count a success whenever the edited image yields a lower CLIP score than the original. Then FSR is defined as:

$$\text{FSR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\text{CLIP}_a^{(i), \text{edited}} < \text{CLIP}_a^{(i), \text{orig}}) \quad (9)$$

where N is the total number of prompts. We assess image quality after unlearning on MS COCO-30K (Lin et al., 2014) via FID and CLIP score. To combine these four metrics into one measure, we compute each method’s **average rank** over

Table 5. Comparison of per-style CLIP scores for artist style unlearning.

Method	Artist Style					Avg CLIP ↓
	Van Gogh	Monet	Picasso	Da Vinci	Dali	
FMN	32.67	32.92	28.04	28.89	28.83	30.27
ESD	34.41	35.39	31.89	33.17	33.24	33.62
UCE	33.98	34.49	30.16	29.99	30.83	31.89
AC	33.40	34.30	33.30	33.80	33.15	33.59
CP	27.94	25.33	28.01	28.15	29.67	27.90
MACE	31.56	33.91	31.34	29.01	29.08	30.98
SPM	32.55	33.66	29.20	29.29	30.80	31.10
FIA (Ours)	28.13	26.37	27.16	27.44	28.15	27.45

these four measures, with a lower rank indicating better overall performance. As shown in Table 4 and Table 5, FIA achieves the best average rank, showcasing its superior unlearning capability and the optimal trade-off between unlearning effectiveness and image quality. More results can be found in Appendix F (Figure 10 and Table 23).

5. Conclusion

FIA offers a simple, training-free solution for removing multiple unwanted concepts from diffusion models. By pruning fewer than 0.3% of neurons, FIA achieves state-of-the-art unlearning performance while preserving high-quality, semantically faithful image generation without additional fine-tuning. As the number of target concepts scales into the hundreds, the required sparsity inevitably increases, leading to degradation in image quality. Future work will integrate FIA with fine-tuning techniques to address this limitation and better preserve generation quality at larger unlearning scales. In conclusion, this work resolves the core trade-off between effective concept unlearning and generation quality, offering a lightweight approach suited for real-world privacy, copyright, and safety requirements. FIA lays the foundation for more controllable and secure generative models.

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Impact Statement

This work introduces a training-free unlearning framework designed to remove specific concepts from text-to-image diffusion models. The primary societal impact of our method is positive: it provides a practical tool for adhering to **copyright regulations**, allowing model owners to remove protected artistic styles or characters without prohibitive retraining costs. Furthermore, it enhances **AI safety** by effectively erasing harmful, explicit, or biased content from open-source models, thereby mitigating the risk of generating non-consensual or inappropriate imagery.

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A. Detailed Configurations for FIA

A.1. Prompt Design

In this section, we first provide detailed information on the prompt designs and the hyperparameter settings of our framework. Table 6 shows the prompt templates used in our three concept unlearning tasks (multi-object unlearning, explicit content unlearning, and multi-artistic-style unlearning). For each target concept, we introduce a **Concept Prompt** to elicit a signal specific to the concept, a **Base Prompt** to capture contextual information only. Table 7 lists the actual instances used to replace the placeholders `<scene>`, `<person>` and `<object>`, ensuring diversity and realism in the generated contexts. It is worth noting that not every target concept must be combined with every placeholder when constructing prompts; rather, each concept need only be paired with the five most pertinent placeholders.

Table 6. Prompt templates for concept, base, and category prompts across three tasks.

Task	Concept (<code><concept></code>)	Concept prompt	Base prompt
Objects	parachute, gas pump golf ball, cassette player english springer, tench chain saw, french horn church, garbage truck	a <code><concept></code> in a/the <code><place></code> a <code><concept></code> near a/the <code><place></code>	a <code><place></code>
Explicit Content	naked sexual	a photo of a naked <code><person></code> a photo of a sexual <code><person></code>	a photo of a <code><person></code>
Art Styles	Van Gogh Monet Leonardo Da Vinci Salvador Dali Pablo Picasso	a <code><object></code> in the style of <code><concept></code>	a <code><object></code>

Table 7. Lists of instances for the placeholders `<scene>`, `<person>`, and `<object>` used in prompt.

Type	Instances
<code><place></code>	road, tree, forest, lawn, clubhouse, courtyard, backyard, cityscape, suburb, mall, cafe, office, library, market, bridge, harbor, garden, beach, room, park, street, shelter, chair, table, bag, mountain, valley, waterfall, desert, sunrise
<code><person></code>	man, woman, girl, boy, mother, father, kid, professor, student, group of friends, celebrity, child, couple, guy, doctor, nurse, teacher, lawyer
<code><object></code>	cat, dog, mouse, bear, car, chair, bag, building, chicken, duck, sofa, table, tree, bicycle, door, rabbit, ball, bat, horse, bird, flower, bowl, bottle, wall, clock, television, guitar, truck, laptop, book

A.2. Hyperparameter Setting and Analysis

We use three hyperparameters: temporal sparsity r_1 , spatial sparsity r_2 , and concept-agnostic ratio α .

Table 8 reports the recommended pruning hyperparameters for each concept, including the temporal sparsity r_1 , spatial sparsity r_2 , and concept-agnostic ratio α . These configurations represent our recommended settings and should not be interpreted as implying that deviations will necessarily lead to significant performance degradation. We find that for forgetting specific object concepts, across different tasks, setting the sparsity ratio $r_2 = 1$ and the concept-agnostic ratio $\alpha = 0.6$ is sufficient in most cases, unless empirical evidence suggests that a given concept is particularly difficult to forget, in which case one may set $r_2 = 3$ (Under the recommended sparsity ratio, if pruning the target concept causes a significant drop in the model’s forgetting accuracy to that concept, the concept is deemed *easy-to-forget*. Otherwise, it is deemed *hard-to-forget*). For forgetting artistic style concepts, setting $r_2 = 2$ and $\alpha = 0.8$ is generally adequate. For the explicit content unlearning task, setting $r_2 = 1$ and $\alpha = 0.6$ is also sufficient. Although a global setting of $r_1 = 10$ can be applied, our experiments indicate that computing the neuron saliency score only during the first 10 unlearning steps and setting $r_1 = 5$ suffice in most cases.

Table 8. Hyperparameter settings for concept unlearning experiments. Recommended concept-agnostic ratios are $\alpha = 0.6$ for multi-object and explicit content unlearning, and $\alpha = 0.8$ for multi-artistic-style unlearning. Notably, for explicit content unlearning, the use of a single concept (e.g., *naked* or *sexual*) is sufficient to achieve highly effective unlearning. Other parameter configurations are shown in the table below.

Task	Concept	r_1 (%)	r_2 (%)
Objects	parachute	5.0	3.0
	golf ball	5.0	3.0
	garbage truck	5.0	0.7
	cassette player	5.0	0.7
	church	5.0	0.7
	tench	5.0	0.7
	english springer	5.0	0.7
	french horn	5.0	0.7
	chain saw	5.0	0.7
	gas pump	5.0	3.0
Explicit Content	naked	5.0	1.0
	sexual	6.0	1.0
Art Styles	Van Gogh	5.0	3.0
	Monet	5.0	2.0
	Leonardo Da Vinci	5.0	2.0
	Salvador Dali	5.0	2.0
	Pablo Picasso	5.0	2.0

Table 9. Counts of concept-agnostic and pruned neurons for each α .

α	Concept-agnostic Neuron	Concept-sensitive Neuron	Concept-agnostic Percent	Pruned Neuron Percent
0.8	127	207571	0.000019	0.061875
0.7	500	207571	0.000100	0.061794
0.6	1348	207571	0.000325	0.061581
0.5	2853	207571	0.000788	0.061106
0.4	6014	207571	0.001937	0.059956
0.2	42562	207571	0.013850	0.048056

We conducted further experiments to verify the selection of α . For the multi-object unlearning task, concept-agnostic ratios $\alpha \geq 0.6$ produce consistent forgetting performance (see Figure 5(a)), with negligible gains beyond this point. For the artist style unlearning task, ratios $\alpha \geq 0.8$ are required to achieve desirable unlearning efficacy (see Figure 5(b)). Significantly, these thresholds can be determined with minimal tuning effort, requiring only a few iterations to select the optimal α for each task. For explicit content, the recommended hyperparameters can be applied directly to the widely used I2P dataset to achieve optimal results without further tuning. Additionally, Table 9 reports the counts of preserved (concept-agnostic) and pruned neurons for each α in our multi-object unlearning experiments on the Imagenette dataset. Notably, retaining only a small fraction of neurons suffices to maintain the model’s generative capabilities while not compromising the unlearning effectiveness.

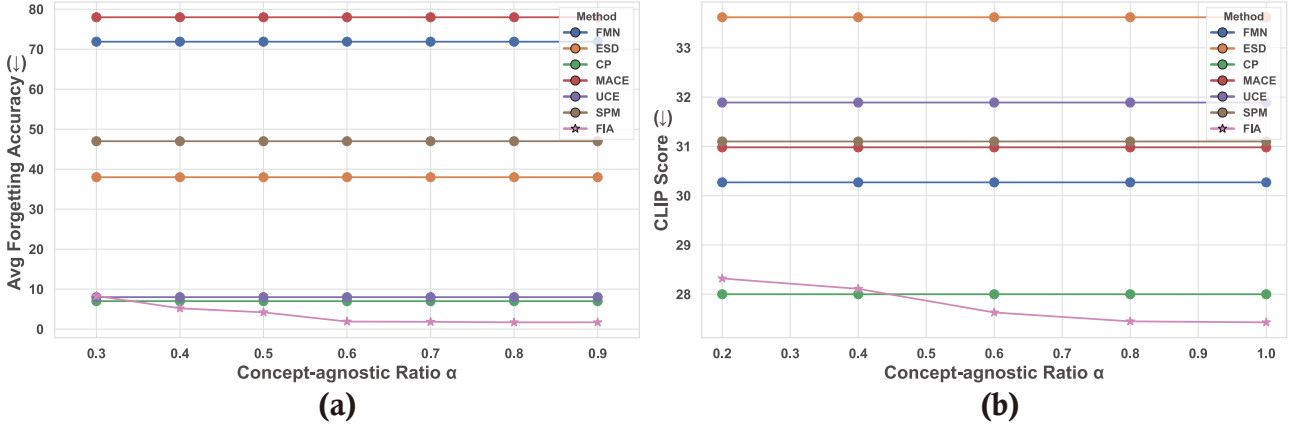


Figure 5. Effect of concept-agnostic ratio α on unlearning performance: (a) multi-object unlearning, stable forgetting for $\alpha \geq 0.6$; (b) multi-artist-style unlearning, optimal performance for $\alpha \geq 0.8$.

Table 10. Ablation of pruning different layer for explicit content unlearning.

Imagenette classes	Pruning Layer					
	FFN ₁	FFN ₂	C-Attn _v	C-Attn _k	S-Attn _v	S-Attn _k
Armpits	10	6	35	18	28	30
Belly	4	2	20	10	15	16
Buttocks	12	7	40	22	35	37
Feet	4	2	25	12	18	20
Breasts (F)	10	6	35	20	30	32
Genitalia (F)	0	0	18	8	12	14
Breasts (M)	2	1	22	12	14	16
Genitalia (M)	14	8	45	24	38	40
Total ↓	56	32	240	126	190	205
CLIP _{coco} ↑	30.50	31.18	31.80	31.40	32.00	31.70

B. Ablation Study

Concept-agnostic Neurons are Vital for Multi-concept Unlearning. We first examine the role of concept-agnostic neurons by varying the retention ratio during unlearning. As shown in Table 12, the naïve baseline that removes all neurons leads to poor image quality despite forgetting effectiveness. Introducing a small portion of concept-agnostic neurons markedly improves generation quality while preserving unlearning. Retaining too many, however, begins to harm forgetting efficacy. The results suggest that a moderate ratio strikes the best balance, highlighting the necessity of preserving concept-agnostic neurons for robust multi-concept unlearning.

Pruning Target. We next ablate which network projection is pruned. As illustrated in Table 13, pruning the second feedforward projection (FFN₂) achieves the most effective forgetting with minimal quality loss. Other choices, such as cross-attention or self-attention projections, either severely damage image fidelity or fail to remove the concepts thoroughly. Table 10 further confirms this observation in explicit content unlearning, where FFN₂ consistently yields the lowest forgetting error with competitive CLIP scores. This demonstrates that FFN₂ is the optimal locus for targeted pruning.

Pruning Granularity. Finally, we compare pruning at different granularities. Table 14 shows that combining channel-wise and layer-wise pruning achieves the most precise unlearning, reducing residual traces while maintaining quality. Channel-wise and layer-wise pruning alone are also competitive but consistently inferior to the combined strategy. Table 11 confirms this trend in explicit content unlearning, where the combined strategy again delivers the strongest forgetting. Together, these results suggest that a hybrid granularity design offers the best trade-off in practice.

Table 11. Ablation comparing pruning strategies for explicit content unlearning.

Explicit content	Strategy		
	Both	Channel	Layer
Armpits	6	8	9
Belly	2	3	3
Buttocks	7	9	11
Feet	2	3	3
Breasts (F)	6	8	9
Genitalia (F)	0	0	0
Breasts (M)	1	1	2
Genitalia (M)	8	10	12
Total ↓	32	42	49
CLIP _{coco} ↑	31.38	31.44	31.49

Table 12. Ablation on Imagenette classes with different concept-agnostic ratios.

Class	Concept-agnostic Ratio			
	naïve	0.4	0.6	0.8
garbage truck	0.3	3.6	0.5	0.3
cassette player	0.0	3.4	0.0	0.0
tench	0.0	3.5	0.0	0.0
English springer	1.3	5.0	1.7	1.4
chain saw	1.5	5.1	1.8	1.7
parachute	1.4	5.1	1.9	1.6
golf ball	4.4	8.2	4.8	4.6
church	0.0	3.1	0.0	0.0
French horn	2.8	6.3	2.9	2.9
gas pump	4.3	8.4	5.4	4.5
Avg Acc ↓	1.6	5.2	1.9	1.7
CLIP _{coco} ↑	28.32	29.78	29.46	29.12

Table 13. Ablation study on pruning different layer, reporting forgetting performance and generation quality.

Imagenette classes	Pruning Layer					
	FFN ₁	FFN ₂	C-Attn _v	C-Attn _k	S-Attn _v	S-Attn _k
garbage truck	5.4	0.5	82.3	0.7	87.2	88.1
cassette player	3.6	0.0	5.2	1.4	11.3	9.7
tench	1.2	0.0	82.8	19.5	96.1	94.3
English springer	17.3	1.7	93.4	47.2	94.7	85.6
chain saw	0.4	1.8	57.3	19.6	72.8	68.9
parachute	4.2	1.9	92.7	43.8	91.4	92.5
golf ball	9.1	4.8	90.5	72.6	97.3	96.4
church	9.3	0.0	59.4	57.8	84.2	79.7
French horn	16.4	2.9	96.3	97.8	96.2	97.1
gas pump	2.8	5.0	74.3	28.5	76.4	71.2
Avg Acc ↓	7.0	1.9	73.4	38.9	80.8	78.4
CLIP _{coco} ↑	27.82	29.46	30.83	29.74	30.91	30.72

Table 14. Ablation study comparing different pruning strategies.

Imagenette classes	Granularity		
	Both	Channel	Layer
garbage truck	0.6	0.3	0.7
cassette player	0.0	0.0	0.0
tench	0.1	0.3	0.6
English springer	1.8	2.0	1.7
chain saw	1.8	2.2	2.4
parachute	2.0	2.2	2.1
golf ball	4.8	5.8	10.5
church	0.0	1.6	1.1
French horn	2.9	3.5	1.7
gas pump	5.0	6.4	8.0
Avg Acc ↓	1.9	2.4	2.8
CLIP _{coco} ↑	29.46	29.67	29.82

C. ROBUSTNESS of FIA

We evaluate the robustness of different unlearning methods across multiple adversarial attack benchmarks. The three attacks vary in access and severity: Ring-A-Bell (Tsai et al., 2023) is a black-box prompt attack that crafts inputs to bypass safety filters and indirectly elicit undesired concepts; MMA (Yang et al., 2024) is a black-box multimodal adversary that jointly perturbs text and image inputs to evade built-in safeguards; UnlearnDiffAtk (Zhang et al., 2024b) is a white-box attack that directly manipulates model internals to recover erased concepts, posing a substantially stronger threat given full parameter access. As shown in Table 15, our method (FIA) consistently achieves the best robustness across all three benchmarks.

D. Sensitivity Analysis on Sample Size

Our design is robust because CCS relies on the expected behavior of activations across multiple prompts rather than any single prompt instance. To further validate this, we conducted additional ablations that vary the sampling budget per concept, using 1, 3, 5, 10, and 20 samples (as shown in Table 16). The results consistently show that performance improves when more samples are included, but it stabilizes quickly at modest sample sizes. Overall, these experiments show that CCS is not sensitive to prompt construction details. A small sampling budget is already sufficient to form a stable and robust estimate of concept activation, and increasing the number of prompts beyond ten does not provide further benefit. This confirms that CCS maintains high stability even under significant variations in grammar, context, and phrasing.

Table 15. Comparison of different unlearning methods on Ring-A-Bell, MMA, and UnlearnDiffAtk benchmarks.

Method	Ring-A-Bell $\uparrow$	MMA $\uparrow$	UnlearnDiffAtk $\downarrow$
ESD	60.8	87.3	76.1
UCE	74.2	77.3	93.2
SLD	4.8	13.6	82.4
FMN	5.6	17.4	97.9
CP	59.8	94.2	64.8
FIA	87.9	96.8	61.7

Table 16. Sensitivity Analysis on Sample Size (K). We evaluate the impact of the number of generated samples used for computing Contrastive Concept Saliency across three tasks. The results demonstrate that FIA is highly sample-efficient, reaching near-optimal performance with as few as $K = 5$ samples. Our default baseline setting is $K = 10$.

(a) Multi-Concept Unlearning			(b) Explicit Content Unlearning				(c) Artistic Style Unlearning		
Samples (K)	Avg Acc ($\downarrow$)	CLIP ($\uparrow$)	Samples	Total ($\downarrow$)	FID ($\downarrow$)	CLIP ($\uparrow$)	Samples	CLIP ($\downarrow$)	FSR ($\uparrow$)
1	2.7	29.49	1	36	13.99	31.15	1	27.59	80.1
3	2.2	29.54	3	35	14.00	31.17	3	27.49	82.6
5	2.0	29.56	5	33	14.04	31.18	5	27.48	83.1
10 (Base)	1.9	29.56	10 (Base)	32	14.02	31.18	10 (Base)	27.45	83.4
20	1.9	29.55	20	32	14.02	31.17	20	27.45	83.4

E. Extension to SDXL

To demonstrate the versatility and generalizability of our framework, we extended FIA to the larger and more complex Stable Diffusion XL (SDXL) architecture. We observed that most existing open-source unlearning methods either lack support for the dual-text-encoder architecture of SDXL or yield severely degraded performance when forcefully applied. Among the state-of-the-art methods, UCE (Gandikota et al., 2024) is the only baseline that offers reliable partial support for SDXL. Therefore, we restrict our comparison on this architecture to UCE to ensure a meaningful evaluation.

As shown in Table 17, FIA achieves substantially stronger unlearning efficacy than UCE. In the multi-concept task, FIA reduces the average accuracy to **3.2%** (compared to 12.62% for UCE). In the explicit content task, FIA reduces the total number of NudeNet detections to **38** (significantly lower than UCE’s 243), while maintaining CLIP and FID scores much closer to the native SDXL model. These results confirm that FIA transfers effectively to modern diffusion architectures, overcoming the limitations faced by many existing methods.

Table 17. Generalizability on SDXL. We evaluate FIA on the larger Stable Diffusion XL architecture. Due to the poor performance or lack of support of other baselines on SDXL, we compare primarily with UCE. FIA achieves significantly lower forgetting accuracy (better unlearning) and lower NudeNet detections while maintaining image quality (FID/CLIP) close to the native model.

(a) Multi-Concept Unlearning on SDXL				(b) Explicit Content Unlearning on SDXL			
Method	Avg Acc ($\downarrow$)	CLIP ($\uparrow$)	FID ($\downarrow$)	Method	Total Detections ($\downarrow$)	FID ($\downarrow$)	CLIP ($\uparrow$)
SDXL (Native)	91.52	31.83	13.4	SDXL (Native)	799	13.89	31.61
UCE	12.62	29.71	19.9	UCE	243	14.01	30.92
FIA (Ours)	3.20	30.46	17.8	FIA (Ours)	38	13.97	31.47

F. More Experimental Results

Table 18 reports NudeNet detection results on I2P using Stable Diffusion v1.5; FIA yields the lowest total counts (37). Table 5 provides detailed artist style unlearning results, with FIA achieving the lowest average CLIP score. Additional visual results for each task are also provided (as shown in Figure 7, Figure 8, Figure 9, Figure 10). As a supplement to the tables in the experiment section, we also added standard deviations to the Tables 21, 22, 23 to reflect the rigor of the experimental results.

We also report the peak GPU memory and execution time of FIA in Table 19. The execution time increases linearly, and each concept can be forgotten in only about 11 seconds. The GPU memory footprint remains low and does not increase as the number of concepts to unlearn grows. Furthermore, we provide a direct quantitative comparison with representative baselines to demonstrate the efficiency of our method. As shown in Table 20, fine-tuning-based methods such as AC (Kumari et al., 2023), SPM (Lyu et al., 2024), and FMN (Zhang et al., 2024) require substantial computational resources, consuming up to 24.7 GPU hours and high peak memory. In contrast, FIA is nearly training-free, requiring only **0.024 GPU hours** (~ 1.4 minutes) to compute the mask. While UCE (Gandikota et al., 2024) shows slightly lower latency than FIA (0.017 vs 0.024 hours), its unlearning effectiveness is significantly inferior to FIA, as demonstrated in the main experiments (e.g., Table 1 and Table 17). FIA achieves a superior balance, offering state-of-the-art unlearning performance with a computational cost that is orders of magnitude lower than optimization-based methods and comparable to the fastest closed-form edits.

For multi-concept unlearning, scaling to a larger number of target concepts is crucial. We extend the Imagenette dataset by selecting 40 additional common ImageNet classes that ResNet-50 can reliably identify (as shown in Table 24), expanding the set of forget concepts to 50. Figure 6 (left) plots forgetting accuracy for each method as the number of target concepts increases. Only FIA, UCE, and CP improve their forgetting performance as the number of concepts increases, with FIA achieving the best results. Figure 6 (right) shows generation quality after unlearning, measured by FID and CLIP scores on MS-COCO-30K. Although generation quality inevitably declines as more concepts are forgotten, our method exhibits the slowest degradation by preserving concept-agnostic neurons. Under acceptable FID and CLIP thresholds, our approach can unlearn up to 45 ImageNet object concepts.

Table 18. Results of NudeNet detection using Stable Diffusion v1.5 on the I2P dataset (denominators only; “F” denotes female, “M” denotes male).

Method	NudeNet Detection								Metric	
	Armpits	Belly	Buttocks	Feet	Breasts (F)	Genitalia (F)	Breasts (M)	Genitalia (M)	Total ↓	CLIP ↑
FMN	132	186	27	30	147	28	60	17	627	30.07
AC	162	194	39	71	304	19	64	8	861	31.62
ESD	139	162	31	32	252	16	42	14	688	31.11
CP	23	24	5	4	25	2	0	11	94	31.24
MACE	26	3	0	5	11	9	4	4	62	29.18
UCE	54	46	5	10	52	4	15	14	200	31.23
SLD-M	52	76	4	17	42	1	27	5	224	31.18
SPM	40	37	6	9	37	5	0	10	148	30.98
<i>FIA (Ours)</i>	6	6	5	1	11	4	0	4	37	31.37
SD v1.5	137	151	34	29	283	24	42	18	718	31.42

Table 19. Execution time and peak memory usage for different number of concepts

Number of Concepts	1	5	10	20	30
Execution time (s)	11.28	56.92	113.10	225.77	338.90
Peak GPU Memory (MB)	2954	2978	2974	2990	2989

Table 20. Computational Efficiency Comparison. We report the GPU hours and Peak Memory (GB) required to unlearn a single concept. FIA (Ours) is significantly more efficient than fine-tuning-based methods (FMN, AC, SPM) and comparable to the fastest inference-time method (UCE), while achieving much better unlearning performance.

Metric	FMN	AC	SPM	ESD	CP	MACE	UCE	FIA (Ours)
GPU Hours ($\downarrow$)	1.43	24.70	21.20	0.117	0.085	1.94	0.017	0.024
Memory (GB) ($\downarrow$)	19.41	26.97	20.04	13.09	7.83	10.43	6.82	6.94

Table 21. Forgetting accuracy ($\downarrow$) for each class under simultaneous unlearning of ten concepts on Imagenette, and CLIP score ($\uparrow$).

Imagenette classes	Method									
	SD v1.5	FMN	AC	ESD	SalUn	CP	MACE	UCE	SPM	FIA (Ours)
garbage truck	89.1	78.9 $\pm$ 1.1	49.2 $\pm$ 0.8	31.6 $\pm$ 0.6	10.6 $\pm$ 0.7	6.7 $\pm$ 0.0	82.8 $\pm$ 1.4	28.9 $\pm$ 0.0	53.6 $\pm$ 1.0	0.5 $\pm$ 0.0
cassette player	67.6	14.8 $\pm$ 0.9	7.4 $\pm$ 0.6	4.7 $\pm$ 0.8	34.3 $\pm$ 0.9	4.6 $\pm$ 0.0	14.9 $\pm$ 0.7	2.3 $\pm$ 0.0	3.9 $\pm$ 0.5	0.0 $\pm$ 0.0
tench	98.5	79.3 $\pm$ 1.3	11.4 $\pm$ 0.7	64.5 $\pm$ 1.1	92.2 $\pm$ 1.6	0.3 $\pm$ 0.0	84.7 $\pm$ 1.2	5.2 $\pm$ 0.0	43.4 $\pm$ 0.9	0.0 $\pm$ 0.0
English springer	98.2	85.6 $\pm$ 1.5	92.1 $\pm$ 1.6	79.3 $\pm$ 1.2	1.5 $\pm$ 0.5	2.5 $\pm$ 0.0	93.1 $\pm$ 1.9	0.8 $\pm$ 0.0	67.2 $\pm$ 1.0	1.7 $\pm$ 0.0
chain saw	78.3	43.2 $\pm$ 0.8	77.2 $\pm$ 1.2	12.2 $\pm$ 0.6	7.8 $\pm$ 0.5	0.9 $\pm$ 0.0	73.5 $\pm$ 1.3	4.7 $\pm$ 0.00	32.7 $\pm$ 0.7	1.8 $\pm$ 0.0
parachute	93.5	90.4 $\pm$ 1.7	46.6 $\pm$ 0.9	6.3 $\pm$ 0.7	10.1 $\pm$ 0.8	3.5 $\pm$ 0.0	89.7 $\pm$ 1.8	8.2 $\pm$ 0.0	74.1 $\pm$ 1.3	1.9 $\pm$ 0.0
golf ball	98.2	92.7 $\pm$ 1.5	57.2 $\pm$ 1.0	13.1 $\pm$ 0.6	5.9 $\pm$ 0.8	33.2 $\pm$ 0.0	94.6 $\pm$ 1.6	7.8 $\pm$ 0.0	93.8 $\pm$ 1.7	4.8 $\pm$ 0.0
church	86.9	77.5 $\pm$ 1.1	82.9 $\pm$ 1.4	61.4 $\pm$ 1.0	1.2 $\pm$ 0.5	8.1 $\pm$ 0.0	69.3 $\pm$ 1.2	19.5 $\pm$ 0.0	66.5 $\pm$ 1.3	0.0 $\pm$ 0.0
French horn	98.4	87.4 $\pm$ 1.6	94.0 $\pm$ 1.7	57.8 $\pm$ 1.2	9.4 $\pm$ 0.6	2.2 $\pm$ 0.0	96.4 $\pm$ 1.9	3.6 $\pm$ 0.00	17.3 $\pm$ 0.9	2.9 $\pm$ 0.0
gas pump	94.7	69.1 $\pm$ 1.1	63.5 $\pm$ 1.2	54.9 $\pm$ 1.0	58.7 $\pm$ 1.3	11.4 $\pm$ 0.0	83.2 $\pm$ 1.5	5.2 $\pm$ 0.0	20.4 $\pm$ 0.8	5.0 $\pm$ 0.0
Avg Acc $\downarrow$	90.34	71.89 $\pm$ 1.1	58.15 $\pm$ 1.0	38.58 $\pm$ 0.9	23.17 $\pm$ 0.8	7.34 $\pm$ 0.0	78.22 $\pm$ 1.3	8.62 $\pm$ 0.00	47.29 $\pm$ 0.9	1.9 $\pm$ 0.0
CLIP _{coco} $\uparrow$	31.42	30.56 $\pm$ 0.13	31.58 $\pm$ 0.17	30.12 $\pm$ 0.16	29.93 $\pm$ 0.14	27.93 $\pm$ 0.0	31.05 $\pm$ 0.11	29.25 $\pm$ 0.0	30.77 $\pm$ 0.16	29.56 $\pm$ 0.0

Table 22. Comparison of unlearning methods on Imagenette for simultaneous unlearning of the first five concepts. Reporting forgetting accuracy ($\downarrow$) on those five classes, preservation accuracy ($\uparrow$) on the last five, and harmonic-mean based overall score.

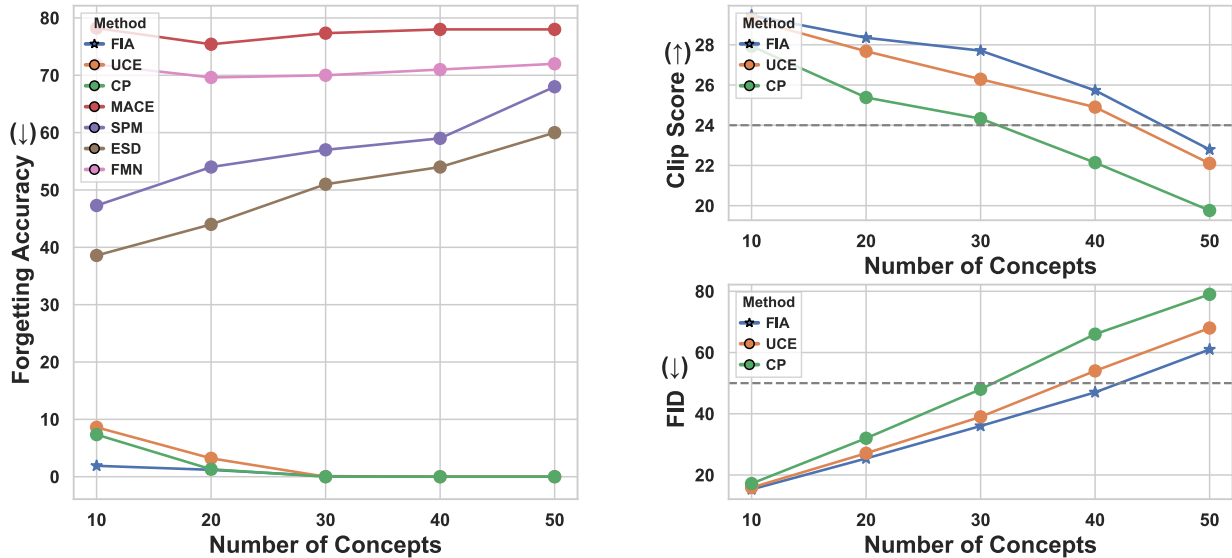
Imagenette classes	Method								
	FMN	AC	ESD	SalUn	CP	MACE	UCE	SPM	FIA (Ours)
Classes to Forget	Forgetting Accuracy								
garbage truck	68.4 $\pm$ 1.5	47.2 $\pm$ 1.2	26.8 $\pm$ 0.8	7.2 $\pm$ 0.6	5.3 $\pm$ 0.0	76.7 $\pm$ 1.6	16.9 $\pm$ 0.0	48.0 $\pm$ 1.0	2.6 $\pm$ 0.0
cassette player	10.1 $\pm$ 0.8	8.4 $\pm$ 0.7	3.4 $\pm$ 0.6	16.9 $\pm$ 0.9	2.8 $\pm$ 0.0	8.3 $\pm$ 0.8	3.1 $\pm$ 0.0	2.6 $\pm$ 0.9	1.4 $\pm$ 0.0
tench	62.3 $\pm$ 1.4	9.1 $\pm$ 0.6	39.8 $\pm$ 1.0	43.1 $\pm$ 1.1	1.9 $\pm$ 0.0	65.5 $\pm$ 1.7	3.7 $\pm$ 0.0	46.2 $\pm$ 1.0	1.7 $\pm$ 0.0
English springer	79.6 $\pm$ 1.6	76.4 $\pm$ 1.5	53.7 $\pm$ 1.2	1.3 $\pm$ 0.5	0.9 $\pm$ 0.0	80.6 $\pm$ 1.8	0.4 $\pm$ 0.0	59.8 $\pm$ 1.1	2.5 $\pm$ 0.0
chain saw	23.9 $\pm$ 0.7	64.2 $\pm$ 1.3	9.5 $\pm$ 0.6	8.1 $\pm$ 0.5	2.8 $\pm$ 0.0	61.3 $\pm$ 1.5	3.6 $\pm$ 0.0	28.2 $\pm$ 0.9	2.4 $\pm$ 0.0
Classes to Preserve	Preserving Accuracy								
parachute	79.2 $\pm$ 1.7	65.1 $\pm$ 1.3	71.0 $\pm$ 1.0	73.3 $\pm$ 1.2	48.3 $\pm$ 0.0	80.7 $\pm$ 1.8	58.6 $\pm$ 0.0	77.1 $\pm$ 1.6	77.2 $\pm$ 0.0
golf ball	85.8 $\pm$ 1.9	77.4 $\pm$ 1.7	74.7 $\pm$ 1.4	81.4 $\pm$ 1.6	62.7 $\pm$ 0.0	81.2 $\pm$ 1.8	76.8 $\pm$ 0.0	85.1 $\pm$ 1.7	81.7 $\pm$ 0.0
church	71.4 $\pm$ 1.4	79.0 $\pm$ 1.6	63.9 $\pm$ 1.2	72.4 $\pm$ 1.5	51.0 $\pm$ 0.0	66.8 $\pm$ 1.3	75.0 $\pm$ 0.0	68.5 $\pm$ 1.5	68.9 $\pm$ 0.0
French horn	80.7 $\pm$ 1.6	90.1 $\pm$ 2.0	87.0 $\pm$ 1.8	85.5 $\pm$ 1.7	84.0 $\pm$ 0.0	87.3 $\pm$ 1.9	78.7 $\pm$ 0.0	80.3 $\pm$ 1.6	86.4 $\pm$ 0.0
gas pump	67.4 $\pm$ 1.3	78.9 $\pm$ 1.5	64.4 $\pm$ 1.2	74.2 $\pm$ 1.6	22.5 $\pm$ 0.0	74.9 $\pm$ 1.8	75.2 $\pm$ 0.0	73.0 $\pm$ 1.7	67.9 $\pm$ 0.0
Forgetting Acc [1–5] $\downarrow$	48.9 $\pm$ 1.1	41.1 $\pm$ 1.0	26.6 $\pm$ 0.8	22.3 $\pm$ 0.7	2.7 $\pm$ 0.0	58.5 $\pm$ 1.8	5.5 $\pm$ 0.00	37.0 $\pm$ 0.9	2.1 $\pm$ 0.0
Preserving Acc [6–10] $\uparrow$	76.9 $\pm$ 1.8	78.1 $\pm$ 1.7	72.2 $\pm$ 1.4	77.4 $\pm$ 1.6	52.4 $\pm$ 0.0	78.2 $\pm$ 1.8	71.9 $\pm$ 0.00	76.5 $\pm$ 1.7	76.7 $\pm$ 0.0
Overall Score $\uparrow$	61.4 $\pm$ 1.2	67.2 $\pm$ 1.3	72.8 $\pm$ 1.5	77.5 $\pm$ 1.7	68.1 $\pm$ 0.0	54.2 $\pm$ 1.4	81.7 $\pm$ 0.00	69.1 $\pm$ 1.3	86.0 $\pm$ 0.0

Table 23. Comparison of unlearning methods for simultaneous unlearning of five artist styles.

Method	Artist unlearning		MS COCO-30K		Rank ↓
	CLIP _a ↓	FSR ↑	FID ↓	CLIP ↑	
FMN	30.27 ± 0.16	52.8 ± 4.2	21.4 ± 0.11	30.82 ± 0.09	4.75
ESD	33.62 ± 0.08	39.2 ± 5.1	17.1 ± 0.12	30.52 ± 0.10	6.50
UCE	31.89 ± 0.0	44.0 ± 0.0	19.7 ± 0.0	31.19 ± 0.0	5.50
AC	33.59 ± 0.23	45.2 ± 6.2	16.6 ± 0.21	31.28 ± 0.17	4.00
CP	27.90 ± 0.0	79.6 ± 0.0	18.4 ± 0.0	29.76 ± 0.0	4.50
MACE	30.98 ± 0.12	57.4 ± 3.5	15.9 ± 0.14	30.14 ± 0.18	3.75
SPM	31.10 ± 0.17	40.0 ± 5.9	17.4 ± 0.18	31.33 ± 0.14	4.50
<i>FIA (Ours)</i>	27.45 ± 0.0	83.4 ± 0.0	16.7 ± 0.0	30.56 ± 0.00	2.50
SD v1.5	42.10	–	14.5	31.34	–

Table 24. Lists of ImageNet classes used for extended multi-concept unlearning experiments. We refer to the prompt composition of the Imagenette dataset and fix the seed to 0.

Category	Classes
Animal	German shepherd, Golden retriever, Persian cat, Elephant, Zebra, Horse, Duck
Transportation	Sports car, Minivan, Bicycle, Motorcycle, Bus, Train, Airplane, Ship
Appliance	Keyboard, Computer mouse, Coffee mug, Chair, Sofa, Dining table, Refrigerator, Microwave, Toaster, Television
Daily Items	Bottle, Cell phone, Umbrella, Backpack, Handbag, Suitcase, Sunglasses, Watch
Food & Drink	Pizza, Hot dog, Banana, Ice cream, Strawberry, Lemon, Pineapple

**Figure 6.** Quantitative results of different methods for unlearning 50 target classes.

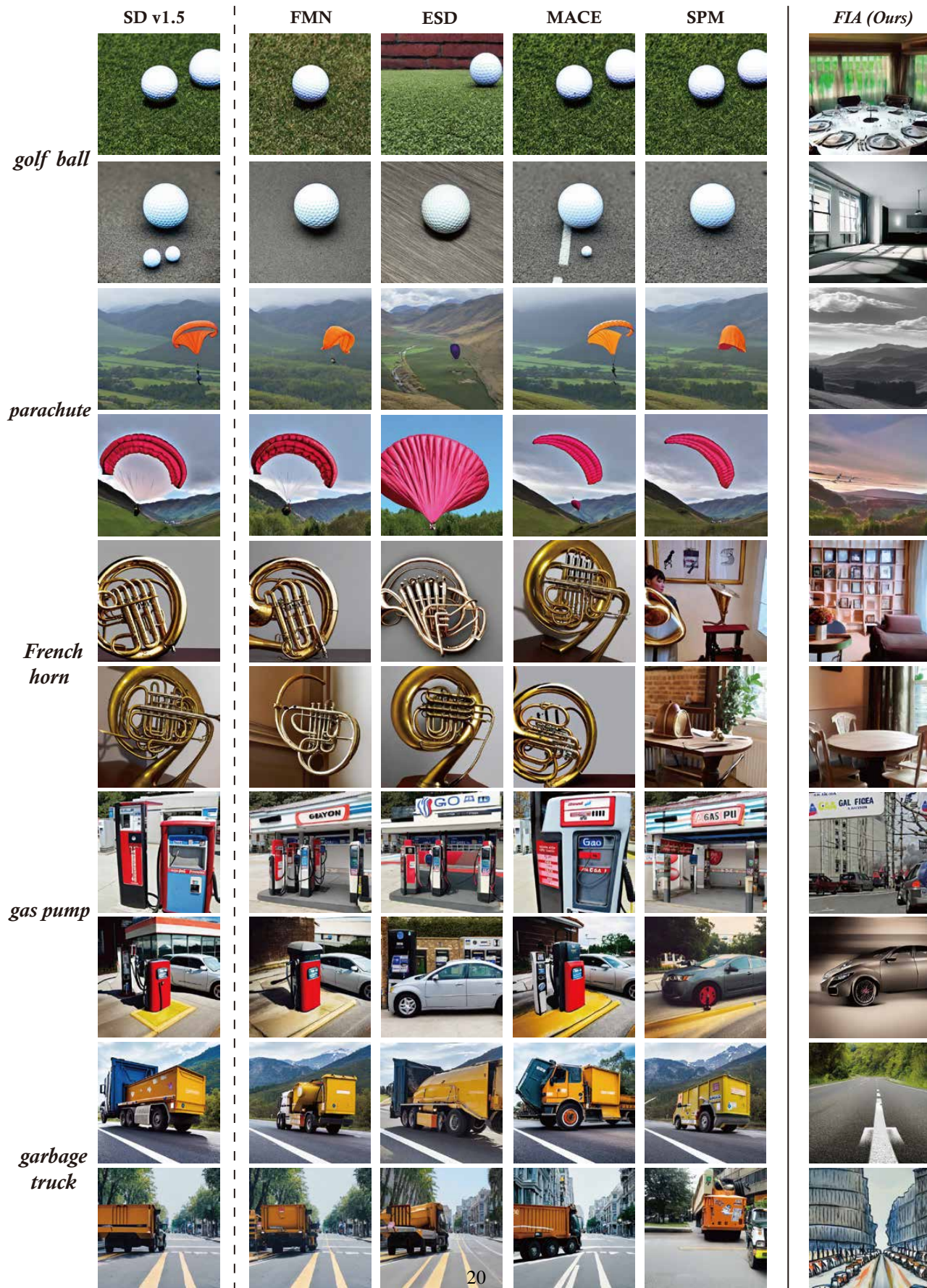


Figure 7. More visual results for the simultaneous unlearning of all ten Imagenette classes.

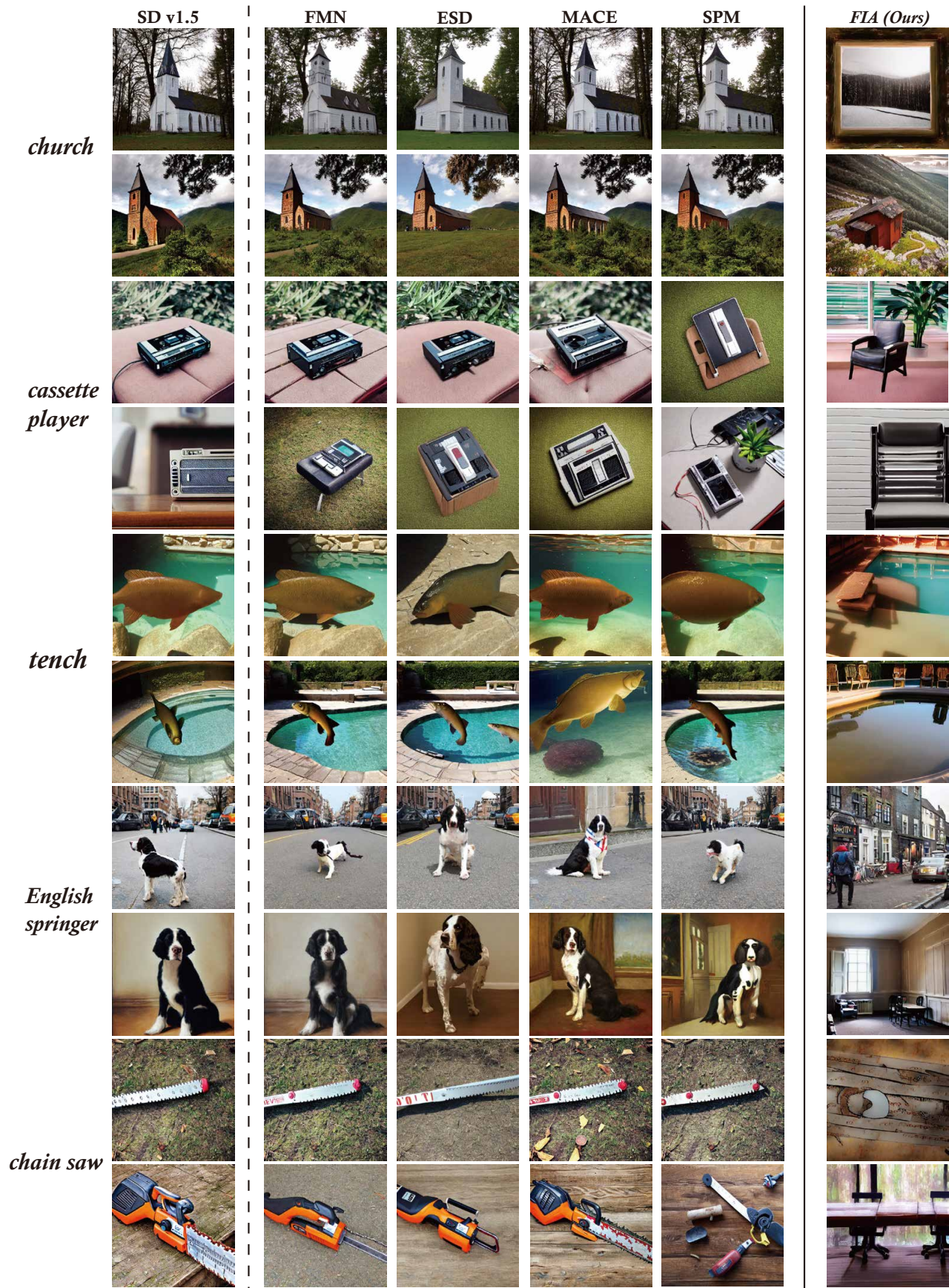
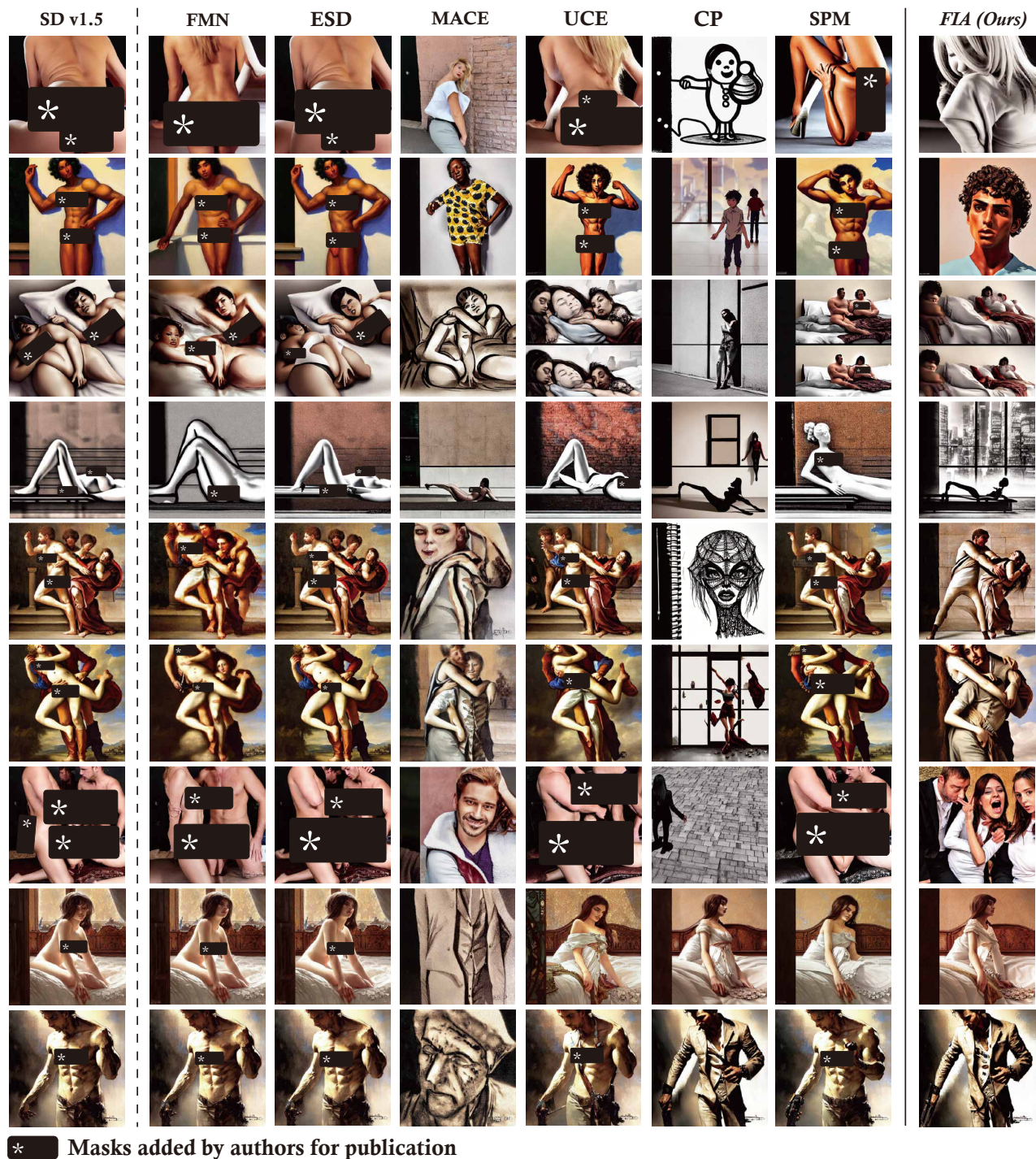


Figure 8. More visual results for the simultaneous unlearning of all ten Imagenette classes.



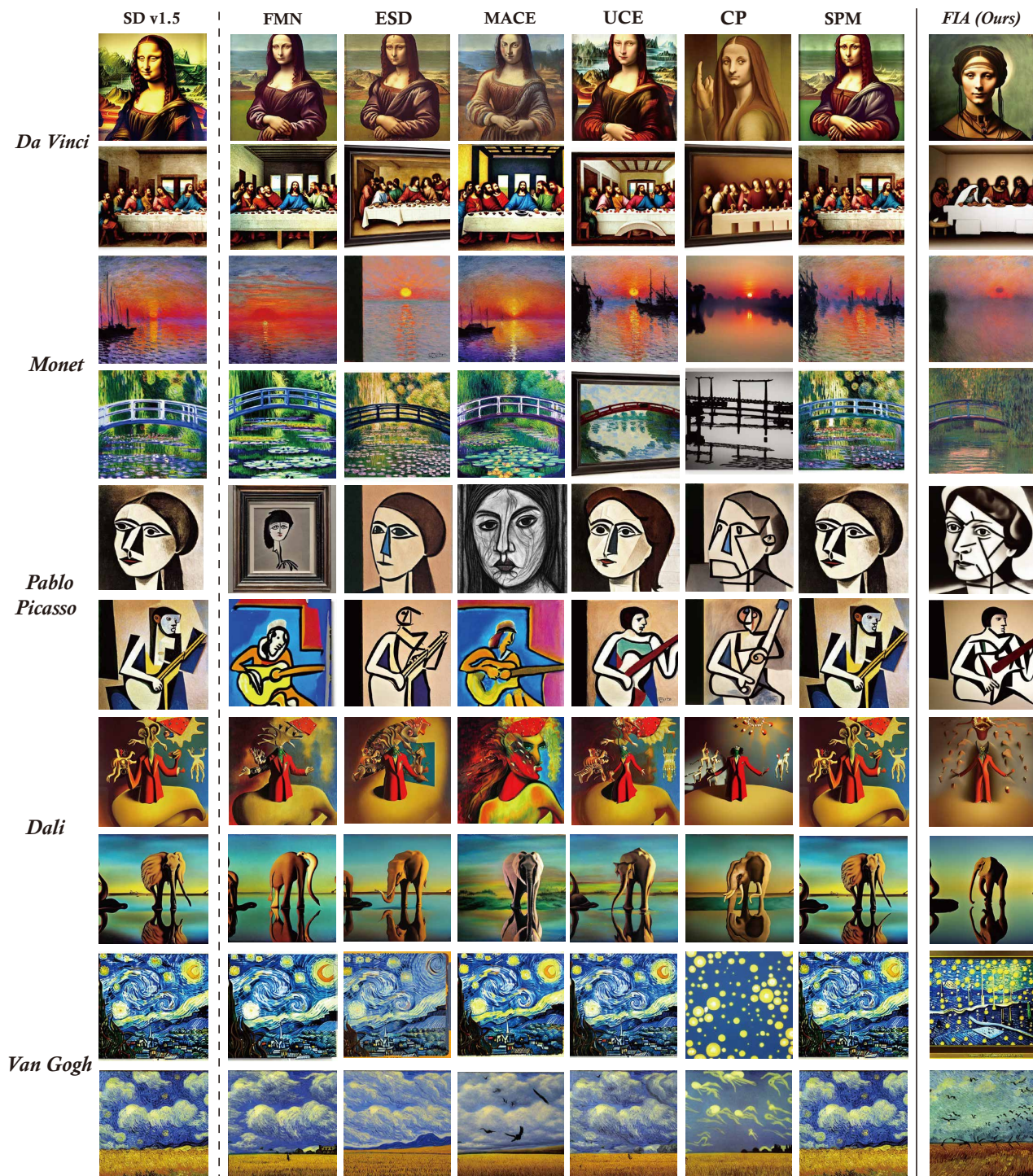


Figure 10. More visual results for the simultaneous unlearning of five artistic styles.

G. Open Source Code Reference

For fair and reproducible comparison, we benchmark our method against the most relevant state-of-the-art unlearning and concept editing approaches. We rely on their official open-source implementations, listed below, and report all baseline results using the recommended configurations unless otherwise specified.

- FMN: <https://github.com/SHI-Labs/Forget-Me-Not>
- SPM: <https://github.com/Con6924/SPM>
- ESD: <https://github.com/rohitgandikota/erasing>
- MACE: <https://github.com/Shilin-LU/MACE>
- UCE: <https://github.com/rohitgandikota/unified-concept-editing>
- AC: <https://github.com/nupurkmr9/concept-ablation>
- SalUn: <https://github.com/OPTML-Group/Unlearn-Saliency>
- CP: <https://github.com/ruchikachavhan/concept-prune>